INCH-POUND
MIL-PRF-83516/4D
16 December 2021
SUPERSEDING
MIL-R-83516/4C
1 November 2011

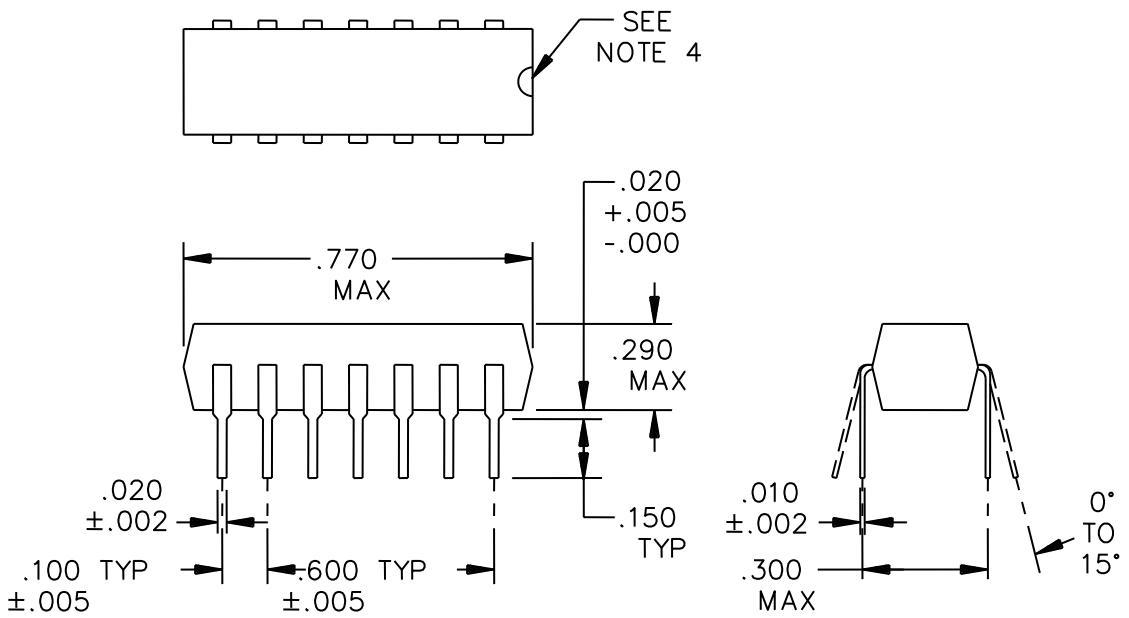
PERFORMANCE SPECIFICATION SHEET

# RELAYS, REED, DRY, DUAL IN-LINE PACKAGE (DIP), GENERAL PURPOSE, LOW-POWER COIL

Reactivated after 16 December 2021 and may be used for new and existing design

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification sheet and MIL-R-83516.



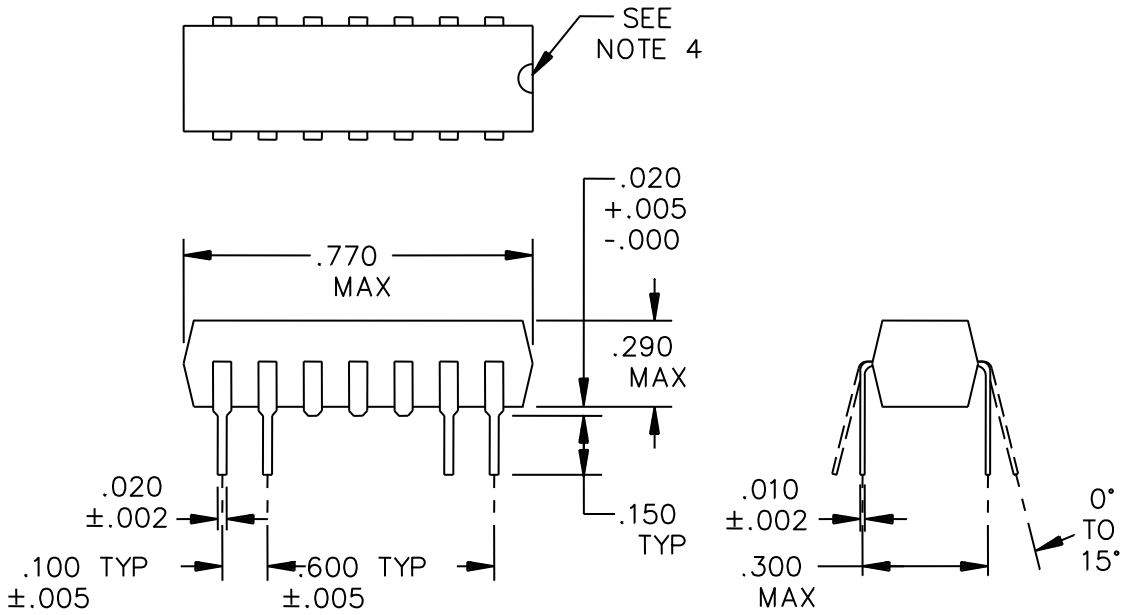

CONFIGURATION A
14 PINS

FIGURE 1. Outline drawing.

AMSC N/A
FSC 5945

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CONFIGURATION B
8 PINS

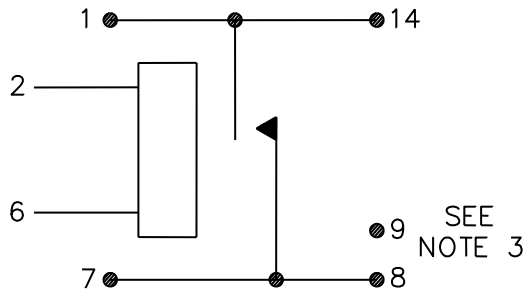
| Inches | mm | Inches | mm |
|---|---|---|---|
| .002 | 0.05 | .150 | 3.81 |
| .005 | 0.13 | .290 | 7.36 |
| .010 | 0.25 | .300 | 7.62 |
| .020 | 0.51 | .600 | 15.24 |
| .100 | 2.54 | .770 | 19.55 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Relays are available with electrostatic shielding at pin 9.
4. A position identifier (shape optional) shall be placed on the top of each relay centered between pins 1 and 14 or directly over pin 1.
5. Relay heights can be found in table I.
6. Unless otherwise specified, tolerance is ±.010 (0.25 mm).

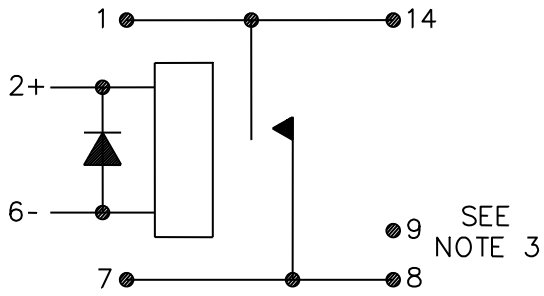
FIGURE 1. Outline drawing - Continued.



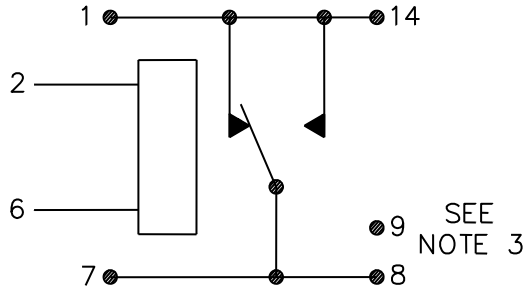
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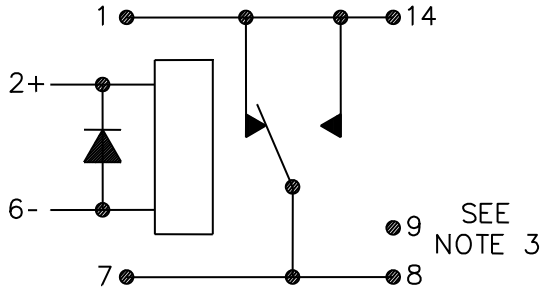

CIRCUIT DIAGRAM A
I FORM A




CIRCUIT DIAGRAM B
I FORM A




CIRCUIT DIAGRAM C
I FORM C




CIRCUIT DIAGRAM D
I FORM C

FIGURE 1. Outline drawing - Continued.



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TABLE I. Dash numbers and applicable characteristics.

| Dash number | | Circuit diagram (See figure 1) | Config-uration (See figure 1) | DC coil data at 25°C | | | |
|---|---|---|---|---|---|---|---|
| Without shielding | With shielding | | | Resistance (minimum) ohms | Pickup voltage (maximum) | Dropout voltage (minimum) | Power dissipation (nominal) mW |
| 5-volt coil | 5-volt coil | | | | | | |
| 001 | 002 | A | A | 450 | 4.0 | 1.0 | 50 |
| 003 | 004 | B | A | " | " | " | " |
| 005 | 006 | A | B | " | " | " | " |
| 007 | 008 | B | B | " | " | " | " |
| 009 | 010 | C | A | 324 | " | " | 170 |
| 011 | 012 | D | A | " | " | " | " |
| 013 | 014 | C | B | " | " | " | " |
| 015 | 016 | D | B | " | " | " | " |

REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figure 1.

Diode characteristics (when applicable): Diodes shall be JANTX as a minimum. 1/

Peak inverse voltage: 75.0 V dc.

Contact data:

Arrangements: See figure 1.

Load ratings:

| Contacts | V dc (max) | mA (max) | VA (max) |
|---|---|---|---|
| Form A | 100 | 500 | 10 |
| Form C | 28 | 250 | 3 |

1/ WARNING: Reverse polarity on coil terminals will destroy diode.



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Contact resistance:

Before life: 0.130 ohm maximum.

During and after life: 1.0 ohm maximum.

Group A inspection: 0.200 ohm maximum.

Contact noise: Applicable.

Contact stability: Applicable.

Coil data: See table I.

Duty rating: Continuous.

Coil power dissipation: See table I.

Operate time: See table II.

Release time: See table II.

TABLE II. Operate and release time (including bounce).

| Contact arrangement | Operate time (nominal) | | Release time (nominal) | |
|---|---|---|---|---|
| | Without diode | With diode | Without diode | With diode |
| | ms | ms | ms | ms |
| Form A | 1 | 1 | .5 | 1 |
| Form C | 2 | 2 | 2.5 | 2.5 |

Electrical data:

Dielectric withstanding voltage: MIL-STD-202-301. 2/

Test points:

Between all mutually insulated components: 500 V dc.

Across contacts: 250 V dc.

2/ For relays supplied with an internal diode, the coil leads shall be connected together to avoid damage to the coil.



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Insulation resistance: 10,000 MΩ minimum.

Test potential: 100 V dc ±10 percent.

Test points: Between contacts.

Coil transient suppression (relays with internal diodes): 1.0 V maximum negative transient.

Capacitance, pF maximum:

| | 1A | 1C |
|---|---|---|
| Across open contacts, no shield: | 1.25 | 2.0 |
| Across open contacts, shield guarded: | 0.25 | 1.2 |
| Contact to coil, no shield: | 3.0 | 3.5 |
| Contact to coil, shield guarded: | 2.0 | 2.0 |

Environmental data:

Thermal EMF: Not applicable.

Thermal shock: Applicable.

Shock (specified pulse): Applicable.

Vibration, high frequency: Applicable.

Salt spray (corrosion): Applicable.

Moisture resistance: Applicable.

Magnetic interference: Relays are subject to interaction with stray magnetic fields. To keep within operating parameters, relays should be mounted no closer than .500 inch (12.7 mm) of each other. Greater spacing is required for stronger magnetic fields.

Resistance to soldering heat: Applicable.

Physical:

Enclosure: Molded plastic.

Operating temperature range: -40°C to +85°C.

Lead integrity: MIL-STD-883, method 2004, test condition B2; the force to be applied shall be 8.0 ±0.5 ounces.

Solderability: Applicable.

Life: Applicable.

Weight: 2.5 grams maximum.

Part or Identifying Number (PIN): M83516/4- (dash number from table I).



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Qualification:

Qualification inspection and sample size: See table III.

TABLE III. Qualification inspection and sample size.

| M83516/4- | Number of samples |
|---|---|
| 008<br>016 | 13 units each part number plus 1 open unit.<br>Qualification inspection as applicable. |

Quality assurance provisions:

Group B inspection not required. Group A inspection required. The qualifying activity shall be notified of any design and/or construction changes and shall impose additional testing requirements as necessary.

MIL-R-83516, paragraph 4.5.1.2.2.1, first sentence does not apply.

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue

Referenced documents. In addition to MIL-R-83516, this document references the following:
MIL-STD-202-301 MIL-STD-883

Custodians:
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2022-006)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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